

Mixing Circular Colourings

Unrefereed candidate proof by GPT-6 Astra

Writeup: gpt-6-astra (single model pass)

Catalog id mixing_circular_colourings_0 — generated 2026-09-16

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Summary and provenance

Catalog id	mixing_circular_colourings_0
Catalog entry	Mixing Circular Colourings
Source corpus	opg
Campaign leg	OpenProblemGarden first pass (attacks_opg)
Source paper / entry	http://www.openproblemgarden.org/op/mixing_circular_colourings_0
Model verdict	proved (confidence: high)
Model's one-line claim	For an n-vertex graph with an edge, the circular mixing threshold is rational, with reduced numerator at most n+1.
Model	gpt-6-astra, reasoning effort max, mode=pro, flex service tier
Original artifact	attacks_opg/mixing_circular_colourings_0/output.md
Independent review	None
Model's caveats	Finite loopless graphs; edgeless graphs require the usual lower-end normalization. Integrality and attainment are not established.

Problem statement

Question Is $\mathfrak{M}_c(G)$ always rational?

Catalog context

Given a proper k -colouring f of a graph G , consider the following ‘mixing process:’

- choose a vertex $v \in V(G)$;
- change the colour of v (if possible) to yield a different k -colouring f' of G . A natural problem arises: Can every k -colouring of G be generated from f by repeatedly applying this process? If so, we say that G is k -mixing. The problem of determining if a graph is k -mixing and several related problems have been studied in a series of recent papers [1,2,4-6]. The authors of [4] provide examples which show that a graph can be k -mixing but not k' -mixing for integers $k' > k$. For example, given $m \geq 3$ consider the bipartite graph L_m which is obtained by deleting a perfect matching from $K_{m,m}$. It is an easy exercise to show that for L_m is k -mixing if and only if

$k \geq 3$ and $k \neq m$. This example motivates the following definition. Definition Define the mixing threshold of G to be

$$\mathfrak{M}(G) := \min\{\ell \in \mathbb{N} : G \text{ is } k\text{-mixing whenever } k \geq \ell\}.$$

An analogous definition can be made for circular colouring. Recall, a (k, q) -colouring of a graph is a mapping $f : V(G) \rightarrow \{0, 1, \dots, k-1\}$ such that if $uv \in E(G)$, then $q \leq |f(u) - f(v)| \leq k - q$. As with ordinary colourings, we say that a graph G is (k, q) -mixing if all (k, q) -colourings of G can be generated from a single (k, q) -colouring f by recolouring one vertex at a time. Definition Define the circular mixing threshold of G to be

$$\mathfrak{M}_c(G) := \inf\{\ell \in \mathbb{Q} : G \text{ is } (k, q)\text{-mixing whenever } k/q \geq \ell\}.$$

Several bounds on the circular mixing threshold are obtained in [3], including the following which relates the circular mixing threshold to the mixing threshold. Theorem (Brewster and Noel [3]) $< /b >]$ For every graph G ,

$$\mathfrak{M}_c(G) \leq \max\left\{\frac{|V(G)| + 1}{2}, \mathfrak{M}(G)\right\}.$$

As a corollary, we have the following: if $|V(G)| \leq 2\mathfrak{M}(G) - 1$, then $\mathfrak{M}_c(G) \leq \mathfrak{M}(G)$. However, examples in [3] show that the ratio $\mathfrak{M}_c/\mathfrak{M}$ can be arbitrarily large in general. Regarding the problem of determining if $\mathfrak{M}_c$ is rational, it is worth mentioning that there are no known examples of graphs G for which $\mathfrak{M}_c(G)$ is not an integer. Other problems are also given in [3]. One can check that $\mathfrak{M}_c(K_2) = 2$ and $\mathfrak{M}(K_2) = 3$. However, the only graphs which are known to satisfy $\mathfrak{M}_c < \mathfrak{M}$ are in some sense related to K_2 , eg. trees and complete bipartite graphs. Question Is there a non-bipartite graph G such that $\mathfrak{M}_c(G) < \mathfrak{M}(G)$? Using an example from [4], it is shown in [3] that if $m \geq 2$ is an integer, then there is a graph G such that $\mathfrak{M}_c(G) = \chi(G) = m$ if and only if $m \neq 3$. A natural question to ask is whether a similar result holds for the circular chromatic number. Again, certain bipartite graphs are an exception. Question Is there a non-bipartite graph G such that $\mathfrak{M}_c(G) = \chi_c(G)$? Also, the example of K_2 shows that the circular mixing threshold is, in general, not attained. However, the following problem is open. Question Is the circular mixing threshold always attained for non-bipartite graphs? For more precise versions of the last three questions, see [3].

Astra writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

A finite set of possible circular mixing thresholds

Write $\mathcal{R}_{k,q}(G)$ for the graph whose vertices are the (k, q) -colourings of G , with two colourings adjacent when they differ at exactly one vertex. Thus mixing means that this graph is nonempty and connected.

For $N \geq 2$, define

$$\mathcal{E}_N = \left\{ \frac{a}{b} : a, b \in \mathbb{Z}_{>0}, \quad 2b \leq a \leq N \right\}.$$

I prove the following stronger statement.

Theorem. Let G be a finite loopless graph with n vertices and at least one edge. Then

$$\mathfrak{M}_c(G) \in \mathcal{E}_{n+1}.$$

In particular, $\mathfrak{M}_c(G)$ is rational, and its reduced numerator is at most $n + 1$.

The main point is that, outside a finite rational set, colourings with a fixed orientation of the edges form a connected reconfiguration class. Connections between these classes can then be tested using systems of difference constraints on at most $n + 1$ variables.

1. Connectivity of integer difference-constraint regions

We first need an elementary lemma.

Lemma 1. Let

$$P = \{x \in \mathbb{Z}^s : 0 \leq x_i \leq K, \quad x_j - x_i \leq c_{ij} \text{ for } (i, j) \in A\},$$

where all c_{ij} and K are integers. Regard A as a directed graph, with arc $i \rightarrow j$ having weight c_{ij} .

Suppose $P \neq \emptyset$, and no directed simple cycle in A has total weight zero. Then the graph on P allowing a change of 1 or -1 in one coordinate at a time is connected.

Proof. First, P is closed under coordinatewise minimum. Indeed, for $x, y \in P$, put $z_i = \min(x_i, y_i)$. If $z_i = x_i$, then

$$z_j - z_i \leq x_j - x_i \leq c_{ij};$$

the other case follows using y .

It suffices to connect x to $z = \min(x, y)$ by successive coordinate decreases. At a current point $x \geq z$, let

$$D = \{i : x_i > z_i\}.$$

A coordinate $i \in D$ can be decreased by one unless there is a tight outgoing constraint

$$x_j - x_i = c_{ij}.$$

The box constraints cannot prevent this decrease because $x_i > z_i \geq 0$.

Moreover, every such blocking coordinate j belongs to D . Otherwise $x_j = z_j$, so

$$z_j - z_i \geq x_j - (x_i - 1) = c_{ij} + 1,$$

contrary to $z \in P$.

If every coordinate in D were blocked, choosing a blocking arc at each vertex of D would produce a directed cycle of tight constraints. Summing around it would give total weight zero, a contradiction.

Consequently some coordinate can be decreased. Integrality ensures that every nontight constraint has sufficient slack for a unit decrease. Repeating reaches z , since $\sum_i (x_i - z_i)$ strictly decreases. Applying the same argument to y completes the proof. $\square$

2. Partitioning colourings by edge orientations

Fix $k, q \in \mathbb{Z}_{>0}$ with $k \geq 2q$. Every (k, q) -colouring f induces an orientation D_f of G : orient uv from u to v when $f(u) < f(v)$.

For an orientation D , let $X_D(k, q)$ be the colourings inducing D . These sets partition the colourings of G . If $u \rightarrow v$ is an edge of D , the corresponding constraints are

$$x_u - x_v \leq -q, \quad x_v - x_u \leq k - q,$$

together with $0 \leq x_v \leq k - 1$.

The variable-to-variable constraint graph therefore has arc weights only of the forms

$$-q, \quad k - q.$$

A directed simple cycle of length b , containing a arcs of weight $k - q$, has total weight

$$a(k - q) + (b - a)(-q) = ak - bq.$$

If this is zero, then $a \geq 1$ and

$$\frac{k}{q} = \frac{b}{a}.$$

Since $b \leq n$ and $k/q \geq 2$, this ratio belongs to $\mathcal{E}_n$.

Lemma 1 consequently gives:

Lemma 2. If $k/q \notin \mathcal{E}_n$, every nonempty $X_D(k, q)$ is connected by single-vertex recolourings. In fact, changes of one unit in the numerical colour suffice within $X_D(k, q)$.

The exclusion of $\mathcal{E}_n$ is important: zero-weight cycles can force several coordinates to move together, preventing single-coordinate connectivity.

3. Feasibility has only finitely many rational breakpoints

We next determine how the nonemptiness of $X_D(k, q)$ can depend on the parameters.

Lemma 3. Fix an oriented graph D on s vertices. The nonemptiness of $X_D(k, q)$:

1. depends only on the ratio k/q , not on its integer representation; and
2. is constant on each open interval in

$$(2, \infty) \setminus \mathcal{E}_s.$$

Proof. Use the difference-constraint graph above, and add a reference vertex $*$, with arcs

$$* \longrightarrow i \text{ of weight } k - 1, \quad i \longrightarrow * \text{ of weight } 0.$$

Setting $x_* = 0$ imposes $0 \leq x_i \leq k - 1$.

This integer difference-constraint system is feasible if and only if the augmented graph has no negative directed cycle. For completeness, sufficiency follows by taking shortest-path distances from $*$. All vertices are reachable; without a negative cycle, these distances are finite integers and satisfy every constraint. The arcs incident with $*$ ensure that the distances lie between 0 and $k - 1$.

It is enough to check simple directed cycles. There are two kinds.

- A cycle avoiding $*$ has weight

$$ak - bq, \quad 1 \leq b \leq s, \quad 0 \leq a \leq b.$$

If $a = 0$, it is always negative. Otherwise, nonnegativity is equivalent to

$$\frac{k}{q} \geq \frac{b}{a}.$$

- A cycle containing $*$ uses exactly one arc of weight $k - 1$ and one arc of weight 0. Its weight is

$$ak - bq - 1, \quad 0 \leq b \leq s - 1, \quad 1 \leq a \leq b + 1.$$

Since $ak - bq$ is an integer,

$$ak - bq - 1 \geq 0 \iff ak - bq > 0 \iff \frac{k}{q} > \frac{b}{a}.$$

Thus feasibility is determined by finitely many weak or strict comparisons of k/q with fixed rational numbers. Every comparison point lying in $[2, \infty)$ belongs to $\mathcal{E}_s$. This proves both assertions. $\square$

The integrality step

$$ak - bq - 1 \geq 0 \iff ak - bq > 0$$

is what removes any apparent dependence on the scale of the pair (k, q) .

4. Testing transitions between orientation classes

For fixed k, q , construct a finite graph $Q_{k,q}(G)$ as follows.

- Its vertices are the orientations D for which $X_D(k, q) \neq \emptyset$.
- Two distinct orientations D, D' are adjacent if some colouring in $X_D(k, q)$ and some colouring in $X_{D'}(k, q)$ differ at exactly one vertex.

We can test each possible adjacency using another oriented graph, now with $n + 1$ vertices.

Suppose the changed vertex is v . Necessarily D, D' agree on every edge not incident with v . Replace v by two nonadjacent twins v_0, v_1 , each having the original neighbourhood of v . Orient:

- the edges at v_0 as in D ;
- the edges at v_1 as in D' ;
- all other edges in their common direction.

A colouring respecting this orientation specifies the unchanged colours on $V(G) \setminus \{v\}$, the old colour at v_0 , and the new colour at v_1 . It therefore gives exactly the desired transition.

Conversely, every such transition yields a colouring of this oriented twin graph. Because $D \neq D'$, at least one incident edge is oppositely oriented, so the two twin colours are automatically different.

Thus adjacency in $Q_{k,q}(G)$ is determined by finitely many feasibility tests of the type in Lemma 3, on $n + 1$ vertices.

When $k/q \notin \mathcal{E}_n$, Lemma 2 says that all nonempty orientation classes are connected. Hence

$$\mathcal{R}_{k,q}(G) \text{ is nonempty and connected} \iff Q_{k,q}(G) \text{ is nonempty and connected.}$$

Indeed, a connected quotient can be lifted by moving within each class to the endpoint of the next transition; the converse follows by projecting a reconfiguration path.

We have proved the central stability statement.

Proposition 4. Let I be a component interval of

$$(2, \infty) \setminus \mathcal{E}_{n+1}.$$

Then either G is (k, q) -mixing for every pair of positive integers with $k/q \in I$, or it is (k, q) -mixing for none of those pairs.

Proof. Throughout I , Lemma 3 makes every vertex-feasibility and transition-feasibility test defining $Q_{k,q}(G)$ constant. Thus the finite graph $Q_{k,q}(G)$ is unchanged.

Also $I \cap \mathcal{E}_n = \emptyset$, so all its nonempty orientation classes are connected by Lemma 2. The quotient criterion now proves the assertion. $\square$

No assertion of scale invariance at the exceptional ratios is needed.

5. A self-contained upper bound

We also need to know that all sufficiently large ratios are mixing. The same machinery gives

$$G \text{ is } (k, q)\text{-mixing whenever } k/q \geq n + 1.$$

Indeed, the palette

$$C = \{0, q, 2q, \dots, nq\}$$

has $n + 1$ colours, and every two distinct palette colours satisfy the (k, q) -constraint when $k \geq (n + 1)q$.

For any colouring f , its induced orientation D_f is acyclic, since colours strictly increase along directed edges. A topological ordering therefore gives an injective palette colouring belonging to $X_{D_f}(k, q)$. Because $k/q \geq n + 1$ lies outside $\mathcal{E}_n$, Lemma 2 connects f to this palette colouring.

Finally, all injective maps from $V(G)$ into C are connected by single-vertex moves that preserve injectivity. To reach a target injective map, fix vertices one at a time. If a desired colour is occupied by another, not-yet-fixed vertex, first move that vertex to an unused palette colour. An unused colour always exists because there are $n + 1$ palette colours and only n vertices.

All these intermediate maps are proper circular colourings. This proves the upper bound, including nonemptiness.

6. Extracting the threshold

Define the set of bad ratios by

$$B_G = \left\{ r \in \mathbb{Q}_{\geq 2} : \begin{array}{l} \text{there exist } k, q \in \mathbb{Z}_{>0} \text{ with } k/q = r \\ \text{such that } G \text{ is not } (k, q)\text{-mixing} \end{array} \right\}.$$

The existential quantifier accommodates possible dependence on the integer representation at exceptional ratios.

Because G has an edge, it is not $(2, 1)$ -mixing: either it has no proper 2-colouring, or the colours on its nontrivial components cannot change. Thus $2 \in B_G$. Section 5 shows that B_G has no element at least $n + 1$.

By Proposition 4, on every component interval I of

$$(2, \infty) \setminus \mathcal{E}_{n+1},$$

either

$$B_G \cap I = \mathbb{Q} \cap I \quad \text{or} \quad B_G \cap I = \emptyset.$$

At the finitely many points of $\mathcal{E}_{n+1}$, arbitrary behaviour is possible and harmless.

Consequently B_G is a union of selected rational open intervals between consecutive points of $\mathcal{E}_{n+1}$, together with a subset of $\mathcal{E}_{n+1}$. It follows that

$$\sup B_G \in \mathcal{E}_{n+1}.$$

Finally,

$$\mathfrak{M}_c(G) = \sup B_G.$$

Every rational $\ell > \sup B_G$ satisfies the defining universal mixing condition, while every $\ell < \sup B_G$ fails it. Whether the condition holds at equality does not affect the infimum.

This proves the theorem. $\square$

Scope and exceptional cases

The proof does **not** assume that every orientation class is connected. That can fail at ratios in $\mathcal{E}_n$, where zero-weight cycles force coordinate differences. Those cases are covered by retaining all exceptional ratios as possible bad points; their detailed behaviour cannot create an irrational supremum.

The argument establishes neither that the threshold is always an integer nor that it is attained.

For edgeless graphs, every colouring reconfiguration graph is connected. Their threshold is therefore the chosen lower endpoint under the usual normalization—for example, 2 if thresholds are restricted to $[2, \infty)$. If the displayed definition is read literally with arbitrarily negative ℓ , an edgeless graph instead gives $-\infty$; that is a trivial normalization issue in the quoted statement, separate from the substantive rationality question.